Azhar Amir Kimanje

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EDUCATION

Seattle Pacific University — **Seattle, WA, USA**
Bachelor of Arts Computer Science and Engineering — *September 2020 - June 2024*

CURRENT RESEARCH & DEVELOPMENT

TCCC SBIR Phase II and IIE - Lead Interoperability Researcher — **Principal Investigator: Dr. Kyle Couperus**
Exonicus Inc, Seattle, WA — *June 2024 - May 2026*

* Led research efforts in developing a real-time XR medical simulation system, focusing on improving data interoperability and optimizing performance for large datasets. This project honed my ability to design data-oriented architectures, conduct rigorous testing, and collaborate with medical experts to ensure accurate simulation outcomes. The research involved innovative approaches to real-time medical data transmission, reducing latency and enhancing system reliability, which contributed to better simulation results for military training.

CourseLLM - Principal Investigator — **Principal Investigators: Azhar Amir Kimanje**
AVELA - A Vision for Engineering Literacy and Access Non-profit, Seattle, WA — *June 2025 - September 2025*

* Led research development for educational support large language model training frameworks for data sovereignty. Led a team of 3 PhD candidates and 6 interns in fine tuning hardware constrained large language models utilizing the Apple MLX framework. Delivered a standalone data activism class lesson plan backed by a specialized large language model to support STEM course progression in partnership with Apple CEI.

Gaussian Splats for Story telling - Principal Investigator — **Principal Investigators: Azhar Amir Kimanje**
AVELA - A Vision for Engineering Literacy and Access Non-profit, Seattle, WA — *January 2025 - August 2025*

* Architected a work flow for utilizing Gaussian splats for story telling in XR utilizing Deep Learning training of 360 drone video to create true to depth 3D environments. In collaboration with Bellevue XR Labs, I published the resources and files under the Washington Open Common Educations Licence.

Avatar Controller Toolkit - ISIA Lab - Research Contributor — **Principal Investigator: Dr. Kevin El Haddad**
Université de Mons., Haunat, Belgium — *August 2024 - Current*

* Contributing to developing a toolkit to visualize and animate facial expressions in XR, contributing to research in computer vision and human-computer interaction. This project allowed me to apply both technical development skills and research methodologies to improve the accuracy and efficiency of facial landmark detection in real-time.

BioDepot-Workflow-Builder - Morphic Lab - Research Contributor — **Principal Investigator: Dr. Ka Yee Yeung**
University of Washington, Tacoma, WA — *September 2024 - Current*

* Contributing to designing a tool to increase reproducibility of gene research in the Morphic Lab. The Bwb GUI is designed for non-programmers who want to use a workflow on their own data without worrying about installation and reproducibility. Bwb auto-installs the components and provides a very simple and intuitive GUI interface for modifying key parameters and accepting user files. We realize that for many users, interactive visualization is an integral part of their analyses. Bwb supports Jupyter notebooks, Cytoscape and other software that have their own graphics and GUIs.

WordPlay - LCI Lab - Research Contributor — **Principal Investigator: Dr. Amy J. Ko**
University of Washington, Seattle, WA — *September 2023 - Current*

* Wordplay is a research initiative led by Professor Amy J. Ko at the University of Washington Information School, focused on justice-centered educational programming. It offers an accessible, multilingual coding platform for creating interactive typography that celebrates global diversity. Designed for diverse secondary classrooms, Wordplay emphasizes inclusion across abilities, languages, cultures, and socioeconomic backgrounds. The project aims to dismantle societal hierarchies in computing by fostering equitable programming practices and advancing justice through research, teaching, and community collaboration.

PUBLICATIONS

From Toy to Tool: An Investigation on Student AI Use Cases and Effects
AVELA, Seattle, WA — *May 2026 - current*

Tool or Toy? Student Views on ChatGPT in Culturally Responsive Computing Education
AVELA, Seattle, WA — *July 2025 - current*

Gaussian Splats for Story Telling
AVELA, Seattle, WA — *June 2024 - current*

Lessons Learned from Creating a Standard for Medical Simulation Interoperability
Exonicus, Seattle, WA — *June 2024 - current*

LEADERSHIP

Director of Mixed Reality and AI Outreach
AVELA University of Washington, Seattle, WA — *June 2024 - current*
Regional Executive Board Finance Chair
National Society of Black Engineers, Seattle, WA — *June 2024 - current*
Chief Executive Office & Founder
Sablesync LLC, Seattle, WA — *November 2023 - current*
President
National Society of Black Engineers - SPU Chapter, Seattle, WA — *June 2024 - current*
Product Owner Software Engineering Capstone
Seattle Pacific University School of Engineering, Seattle, WA — *September 2023 - June 2024*
Freshman Counseling Team Lead
SPU Undergraduate Admissions Office, Seattle, WA — *April 2023 - June 2024*

ENTREPRENEURSHIP

Director of Mixed Reality Outreach — **$150,000**
AVELA University of Washington, Seattle, WA — *June 2024 - current*
Chief Executive Office & Founder
Sablesync LLC, Seattle, WA — *November 2023 - current*
Regional Executive Board Finance Chair — **$160,000**
National Society of Black Engineers, Seattle, WA — *June 2024 - current*
Principal Investigator — **$66,000**
*AVELA Non profit * CourseLLM Application, Seattle, WA* — *September 2024*

WORK EXPERIENCE

Lead Interoperability Engineer, Cybersecurity Analyst
Exonicus Inc., Seattle, WA — *June 2024 - Current*
· Led a interoperability engineering in a team of **5** integrating **JETS** & MOHSES **API**, designing CI/CD pipelines.
· Spearheaded IEEE quality assurance for game engine plugins for **MR**, allowing interoperability of core-product
· Developed real-time data synchronization and controlled frequency protocols to physiology engines and virtual patients.
· Conducted system testing and debugging, accelerating bug resolution in quest hardware-software communication systems.

Director of Mixed Reality and AI Outreach
AVELA - A Vision for Engineering Literacy and Access, Seattle, WA — *June 2024 - Current*
· Spearheaded full stack development of education simulation and digital twin of the Washington NanoFabrication Facility.
· Facilitated new partnerships to AVELA UW with Apple Education, Washington NanoFabrication Facility, and SPEEA ACE.
· Delivered **XR** curriculum to **3200+** STEM students annually combat data activism and environment awareness.
· Designed pipeline for acquiring **30** quest 3 headsets, **60** latops (Mac and Android), and **30** Ipads for **XR**.

Software Engineer, XR - Design Intelligence
IA Interior Architects Inc., Seattle, WA — *June 2024 - September 2024*
· Verified all **built-environment** XR models using Unity and Autodesk REVIT for fast prototyping and design accuracy.
· Developed 10+ XR space plans using Unity and C# to optimize space utilization.
· Researched AI/GenAI solutions using Python, enhancing firm processes.
· Assisted in cost estimation, reducing project time by 20% through efficient data analysis.

Lead Feature Engineer - Extended Reality Course Designer
AVELA - A Vision for Engineering Literacy and Access Non-profit, Seattle, WA — *November 2023 - June 2024*
· Spearheaded training of conversation AI agents in XR as educational assistants for computing education.
· Introduced full stack development of Augmented Reality applications and 3D model training from 360 images.
· Developed the first **40**-hour XR curriculum for AVELA UW, increasing student STEM engagement by **25%**.

Technical Mixed Reality Lead Instructor
SPEEA Aerospace Career Enhancement (ACE), Tukwila, WA — *December 2024 - Current*
· Delivered **XR** training utilizing computer vision models for Gaussian splats.
· Designed drone-based solutions for spatial planning and agriculture.
· Spearheaded **VR** and **AR** initiatives for aerospace and sustainability projects.
· Conducted system testing and debugging, accelerating bug resolution in hardware-software communication systems.

STEM Lead Instructor
AVELA University Of Washington, Seattle, WA — *September 2023 - November 2023*
· Spearheaded grant writing to increase XR curriculumn intergration and hardware into AVELA UW.
· Taught python programming, data activism, and computing for environmental awareness.

Founder & CEO
Sablesync, Seattle, WA — *November 2023 - June 2025*
· Led AR app development, reducing system integration time by **50%.**
· Managed a **3**-engineer team in real-time hardware communication for AR, cutting development cycles by **40%.**

· Integrated automated testing for AR hardware components, reducing firmware errors.
· Optimized deployment processes using Oculus Integration XML, improving hardware-software efficiency.

Product Owner Software Engineering Capstone - SPU GradPro
Seattle Pacific University, Seattle, WA *September 2023 - June 2024*
· Developed SPU GRAD PRO in Unity and deployed it on WebGL to simulate a 3D undergraduate experience.
· Implemented CI/CD pipelines for streamlined development using GitHub, Jira, and Confluence.

Augmented Reality Development Extern
Snap Inc, Seattle, WA *March 2024 - April 2024*
· Designed augmented reality experiences for sport business using Lens studio Beta 5.5
· Curated product initiation requests and systems design documents to satisfy customers demand and stakeholders.

Product Management Intern
YouthForce Boys and Girls Club, Seattle, WA *November 2023 - Jan 2024*
· Conducted 5 successful beta tests of completed Youthforce XR lounge hosting 30 beta testers at a time.
· Organized data collected and implemented new features upon release of final product to Meta App Labs

Software Engineering Intern
YouthForce Boys and Girls Club, Seattle, WA *July 2023 - September 2024*
· Led full-stack product development, reducing MVP release time by **60%** within **2** months.
· Optimized **3D** graphics rendering from **60** to **120** fps using CPU & GPU profiling and Unity frame debugger.

Freshman Counseling Team Lead
YouthForce Boys and Girls Club, Seattle, WA *April 2023 - June 2024*
· Spearheaded a team of 12 freshman counseling intern making 1200+ outbound increasing university enrollment by 35%.
· Conducted 15 interviews hiring 3 new freshman counseling interns.

Freshman Counseling Intern
YouthForce Boys and Girls Club, Seattle, WA *December 2020 - April 2023*
· Led the freshman team in Outbound call rate making 200+ calls per week while giving school tours.
· Increased admission rate of minority students into the school of engineering by 25% from student engagment.

FAVORITE PROJECTS

SBIR Phase II and IIE - Lead Interoperability engineer
Exonicus Inc, Seattle, WA *June 2024*
· **Developed a custom API plugin**, reducing MVP deliverable time by **11** months for real-time medical data communication.
· Collaborated with medical investigators and product managers to ensure alignment with D.O.D. training requirements.
· Optimized data handling with **Unity DOTS**, processing thousands of medical injuries while minimizing cache misses.
· Integrated interoperability prototype into MiLEXTs, enabling scalable, open-source military training simulations.
· Standardized data categorization and naming conventions, enhancing usability of extensive medical datasets.

Security Digital Twins - Full Stack Developer
Sablesync LLC, Seattle, WA *June 2024*
· Built an emergency evacuation simulator on Unity game engine using Unity AI.
· Utilized Unity Navmesh and Unity Navigation AI to model NPC guidance
· Automated 3D digital twin location creation from floor plan to 3 model using Nvidia Omniverse OmniCode.
· Designed prototype for MVP showcase in 1 month of R&D with a team of 3 engineers

Project Wanderus - Product Manager and BackEnd Developer
Amazon AWS & Inrix, Seattle, WA *December 2023*
· Worked in a team of 3 Engineers on Yelp API integration for personalized restaurant and activity recommendations.
· DALL·E API generates custom visuals for trip itineraries.
· Real-time data analysis curates personalized day plans automatically.
· Interactive map highlights tailored points of interest for easy navigation.

XR Museum - Full Stack Developer
Sablesync LLC, Seattle, WA *September 2023*
· Built XR Museum in C#, JavaScript, on web environment and Oculus Virtual Reality Hardware.
· Optimized performance, achieving up to 350 FPS with 350k batches for tris and verts.
· Designed player physics and mechanics using C# scripting and Unity Inspector tools.
· Optimized performanc**e**, achieving up to 350 FPS with 350k batches for triangles and vertices.

ORGANIZATIONS

Simulation Interoperability Standards Organization
Exonicus, Inc, Seattle, WA *Feb 2025*

Alliance for OpenUSD (AOUSD)
Sablesync, LLC, Seattle, WA *June 2025*

Northwest XR EDU NEtwork (NW XR EDU)
AVELA, Seattle, WA *Feb 2024*

AWARDS

1st Place Top Poster Presentation Award
Seattle Pacific University, Seattle, WA — *June 2024*

1st Place University of Washington Tech Showcase — **$1000**
University of Washington, Seattle, WA — *June 2024*

3rd Place out of 200 Global Eduhacks — **$500**
University of Washington, Seattle, WA — *June 2024*

1st Snap Inc Master of Augmented Reality — **Won Internship Interview**
Snap Inc, Seattle, WA — *April 2024*

6th Place out of 150 Amazon AWS & Inrix Hackathon — **Won Internship Interview**
Amazon University, Seattle, WA — *December 2023*

1st Youthforce Coding Challenge — **$700**
National Boys and Girls Club, Seattle, WA — *July 2023*

GRANTS

Lead Interoperability Engineer — **$750000**
Exonicus, Seattle, WA — *September 2025*

Director of Mixed Reality Outreach — **$75,000**
Apple, Seattle, WA — *September 2024*

Lead Interoperability Engineer — **$1,900,000**
Exonicus Inc, Seattle, WA — *May 2024*

Chief Product Officer — **$66,000**
*AVELA Non profit * Washington NanoFabrication Facility, Seattle, WA* — *September 2024*

Lead Extended Reality Instructor — **$65,000**
University of Washington Technology Fee, Seattle, WA — *February 2024*

Washington Bridge Grant — **$3,000 per year**
Seattle Pacific University, Seattle, WA — *June 2020 - June 2024*

Director's Grant — **$8,000 per year**
Seattle Pacific University, Seattle, WA — *June 2020 - June 2024*

Culture Tech Legacy Google Scholarship — **Fully Paid Certifications**
National Boys and Girls Club, Seattle, WA — *February 2024*

Culture Tech Legacy Master class Scholarship — **Fully Paid Certifications**
National Boys and Girls Club, Seattle, WA — *February 2024*

CERTIFICATES

Microsoft
Extended Reality Development & Cloud Development, Seattle, WA — *February 2024 - June 2024*
Microsoft Azure Digital Twins.
Microsoft Connect IoT data to Mixed Reality with Azure Digital Twins and Unity
Introduction to Azure Digital twins
Mixed Reality and Azure Digital Twins
Microsoft Office Specialist: Excel Associate (office 2019)

Meta
Augmented Reality Development, Seattle, WA — *September 2024*
Meta Spark Foundations of Augmented Reality

Exonicus Inc
Cybersecurity , Seattle, WA — *June 2024*
OPSEC Awareness For Military Member, D.O.D Employees

Snap Inc
Augmented Reality Development, Seattle, WA — *March 2024*
Snap Inc Master of Augmented Reality Development

Unity
AVELA University of Washington, Seattle, WA — *July 2023 - June 2024*
Unity VR development
Unity Shaders and Materials
Unity Data Oriented Design and Entity Components Modeling

Unity Junior Programmer

Google
Cybersecurity, Seattle, WA *March 2024 - April 2024*
Google Connect and Protect: Networks and Network Security
Foundations of Cybersecurity
Google Play it Safe: Manage Security Risks

LinkedIn
Software Development & Cybersecurity, Seattle, WA *December 2023 - June 2024*
LinkedIn CISSP Identity and Access Management
LinkedIn Generative AI for Product Managers
LinkedIn Certificate of Web Programming Foundations
LinkedIn Certificate of C# Development

TECHNICAL SKILLS

Programming Languages: **C#** - 5 years, **C++**, **C**, **SQL**, **Python**, **Java**

Frameworks: .Net, Unity & Unreal game engines, RealityKit, AR-Core, ARKit, three.js, Nvidia Omniverse, OpenXR

Career Interest: Doctor of Philosophy in Computer Science

VOLUNTEER WORK

XR technical Support
National Boys and Girls Club, Seattle, WA *current*

Scholarship Essay Writing Support and Coding Instructor
AVELA University of Washington, Seattle, WA *current*

Welcome Team
Seattle Pacific University, Seattle, WA *November 2023 - June 2024*

Student Volunteer
National Boys and Girls Club, Seattle, WA *current*